

Institute of Actuaries of India

Subject SA3 – General Insurance

April 2016 Examination

INDICATIVE SOLUTION

Solution 1:**i)**

Separate the premium into different risk types and calculate approximate delays between the policy inception date and the date of claim for each of the different risks to calculate the earning pattern for each risk e.g. cancellation charges will begin exposure before the travel start date, whereas other risks will commence on the travel start date.

Combine the different risks in each policy to work out a pattern at the individual policy level

Build up the earning pattern on a 365ths basis from all the individual policies taking into account these earning patterns, inception dates and expiry dates of each policy

[4]

ii)

Data: Policy file details on premium, inception and expiry dates, by type of risk, policy type (including geographical differences), limits of insurance

Monitoring model: Using written premium from data and earned premium calculated as per methodology described above, compare actual premiums (net of expenses) with claims experience as it emerges from the portfolio for health/life cover, property covers (eg loss of luggage), cancellation covers and liability. Split this model structure by significant category of cover (eg International – split US/Rest of World – and Domestic)

[4]

iii)

In principle, annual policies may be treated in the same way as other annual business for the company

Would need to check whether the spread of risk had a particular seasonal weight (using company and/or industry experience), which should be incorporated into the liability measurement model if possible

Again you would need to determine whether the annual business should be broken down by significant risk type

[3]

iv) Advantages:

Increased sales possibility of adding travel to other domestic covers, particularly in relation to annual cover opportunities

Increased cross-selling opportunities

Disadvantages:

Possibility of anti-selection (only customers who take this out may be those who travel a lot)

Underwriters and claims staff may not have the necessary expertise to write these types of risk, which are a combination of property and liability risks

[4]

[15 Marks]

Solution 2:**i) Key issues:**

- Sufficiency of Tautai's insurance liabilities
- Solvency level after the acquisition
- Historical profitability
- Value of new business (future profit and the hurdle rate)
- Impact on expenses after the acquisition

[3]

ii) Risks and risk management strategy:

- Larger exposure to large Motor claims: XOL or surplus reinsurance
- Solvency margin: rights issue/capital injection, slow future premium growth, reduce exposure to equities, reduce retention
- Event exposure from the integrated Motor exposure in the same region: Event RI program.
- Investment return volatility: Reduce exposure to equities and properties

[3]

iii) Factors:

- Integrated rating factors and the data availability
- Historical claims experience
- Rate increase impact on ex- Tautai policies; hence capping and uncapping rate changes for the renewal book
- Additional/different product features for the new harmonised product
- Reinsurance and expense structure of ex-Tautai policies

[4]

iv) Students should be able to clearly see that the loss ratio and renewal rates are vastly different for different components of the portfolio. It should also be obvious that the lapse experience differs between Country and Metro and that the Metro business appears to be quite price sensitive. Key observations:

- Metro renewal rates decline as size of car increases;
- Country renewal rates are the reverse (though not as much change);
- The sectors with the lowest renewal rates tend to have the best loss ratios, which indicates that they are adequately (or over-) priced;
- Students should undertake a range of calculations looking at the potential impact of the rate increase on lapse rates and the loss ratio. If it is assumed that the recent lapse experience is repeated then it will show that the lapse rate could be expected to increase significantly and that the loss ratio will reduce to close to the target level. Calculations of portfolio averages should be weighted by the various exposure measures;
- Comments on the analysis should highlight the key assumptions being made – e.g. that lapse rate experience will repeat, that loss ratio experience will repeat

- Students should note that an across the board increase is unlikely to result in the desired reduction in loss ratio. It would be expected that a higher proportion of the better risks will leave than poorer risks thus reducing the improvement in the loss ratio;
- The recommendation should state that the across the board increase is unlikely to produce the best outcome. A more focused increase in the areas of the business with poor claims experience is likely to produce a better outcome. The recommendation should note the potential impact of a significant increase in lapse rate as well as the potential for the new business attracted to be, on average, worse than recent new;
- Further work that should be suggested should include
 - checking what competitors are doing and monitoring the profile and experience of new business
 - supplement the analysis with experience of Moimoi insurance, making suitable allowance/adjustment as appropriate
 - assessment for and allowance of potential synergy benefits, such as consolidated reinsurance structure, mainframe, product and claim management resource etc

[7]

[17 Marks]**Solution 3:****i)**

Past claims updated to December 31, 2015 (middle of policy year). Estimate price index at December 31, 2015 of 172.

Inflation factors

Loss no	Date	Inflation factor
1	April 2010	1.165
2	August 2011	1.112
3	December 2011	1.104
4	July 2012	1.090
5	June 2013	1.047
6	June 2014	1.031

Notes:

1. December 2015 index (172) estimated.
2. Inflation factor = $172 / \text{index for month}$, eg. $1.104 = 172 / [154.3 + (157.5 - 154.3) \times 11/24]$
3. Inflation factor uses linear interpolation between June indices; solution assume losses occur mid-month - students may use end-of-month or other simplifying assumptions

Allocation of Losses to Policy Year, Indexed Policy Losses and Loss to Policy

Loss No	Month cost unit incepted or renewed	Date of loss	Policy year	Loss severity Rs mllns	Inflation factor	Indexed loss	Loss to policy
1	Nov 2009	Apr 2010	2009	4.00	1.165	4.660	1.660
2	Nov 2010	Aug 2011	2010	2.50	1.112	2.780	0
3	Mar 2011	Dec 2011	2010	5.80	1.104	6.403	3.000
4	Oct 2011	Jul 2012	2011	4.00	1.090	4.360	1.360
5	May 2013	Jun 2013	2012	5.00	1.047	5.235	2.235
6	July 2013	Jun 2014	2013	5.00	1.031	5.155	2.155

Notes:

1. Indexed Loss = Loss Severity x Inflation Factor.
2. Loss to Policy = max(Indexed Loss - Rs 3m deductible, 0), restricted to Rs 3m limit.
3. Two most recent losses are not yet paid and there may be development on loss amount; ignored in this answer but students may add some IBNER factor.
4. Assumes no losses other than those reported; students could allow for IBNR losses.
5. Assumes no more losses in current policy year. Students could allow for future losses in current policy year.
6. Policy year is year beginning 1 July XXXX.

[6]

ii) Policy year exposure

Date	Number of cost units	Policy year exposure
July 1, 2008	10	13
July 1, 2009	16	20
July 1, 2010	24	26
July 1, 2011	28	29
July 1, 2012	30	33
July 1, 2013	35	37
July 1, 2014	38	39
July 1, 2015	40	40

- Note:
1. Policy Year Exposure = [No. cost units 1/7/X + No. cost units 1/7/(X+1)] / 2.
 2. Assumes two more cost units opt in during current policy year.
 3. Policy year begins on date in first column.
 4. July 1, 2015 number (40) estimated.
 5. Do not have to allow for new cost units in upcoming policy year as calculated premium will be used.

Calculation of indexed costs by year

Factor	2010	2009	2010	2011	2012	2013	2014
Losses allocated to year	0	1.666	3.000	1.360	2.235	2.155	0
Exposure Adj Factor – Cost Unit Cost	3.077	2.000	1.538	1.379	1.212	1.081	1.026
Indexed loss cost	0	3.320	4.615	1.876	2.709	2.330	0

Notes:

1. Losses allocated to years = policy year losses from table above.
2. Exposure Adj. Factor - Cost Unit Count = 40 / Policy Year Exposure, eg. $1.538 = 40/26$.
3. Indexed Loss Cost = 1 x 2 , eg. $4.615 = 3.000 \times 1.538$.
4. No adjustment in solution for partially completed policy year.
5. Assumes no losses other than those reported (i.e. no IBNR losses).
6. Policy year is year beginning 1 July XXXX.

[10]

iii) All amounts in Rs millions

Loss No	Gross Loss	Recovery (original cover)	Recovery (1 st reinstatement)	Recovery (2 nd reinstatement)	Premium (original cover)	Premium (1 st reinstatement)	Premium (2 nd reinstatement)	Net loss
1	5	2.5			0.5	0.5		2.5
2	3		0.5				0.1	2.5
3	10		2.0	0.5			0.4	7.5
4	6			2.0				4.0
Total	24	2.5	2.5	2.5	0.5	0.5	0.5	16.5

- Final loss recovery is limited to Rs 2.0m even though the loss exceeds the deductible by Rs 3.5m.
- Aggregate capacity of whole contract is Rs 7.5m.
- No reinstatement premium is payable after the third loss, as the limit will not be reinstated once exhausted.
- Second reinstatement premium in solution has been paid in proportion to cover used (implied by question wording). i.e., 0.1 is paid, as Rs 0.5m (20% x Rs 2.5m) cover is used.

[4]

[20 Marks]

Solution 4:

i)

IoT explanation should cover:

- Network of physical things / objects interconnected

- Enables exchange of data between these networked things / things talking to each other
- Through installed technology / software / sensors etc.

Areas of opportunity:

Products

- Automobile insurance – pay as you drive / telematics – usage based insurance to customers
- Health care – based on health sensors through mobile phones, smart watches or other gadgets
- Contents insurance – usage based insurance using sensors / chips in the electrical and electronic appliances
- Property insurance – through real time data on weather, temperature, seismic activity
- Marine insurance - safety of maritime shipping – connecting the ships at sea – port operations, service providers
- Accidental insurance – wearable technology to monitor hazardous / dangerous working conditions and to analyse behavioural data of workers working in manufacturing sector
- Crop insurance – real time information on weather – insights into field activity – sensors to accurately estimate conditions, wind speed, humidity levels etc. removing the much prevalent 'hunch' dependence on sowing and harvesting
- Business Interruption – real time tracking of supply chain – avoidance / better management of business interruption incidents

Claims & Servicing

- Innovation - ability to predict consumer behaviour and trend – using Internet and mobile device usage
- Value added services tailor made for customers

[6]

ii) a)

- i) Data requirements & monitoring:
 - a. Initial data on member activity and fitness level, reading of vital statistics like BMI, pulse rate, blood pressure
 - b. Ongoing monitoring of member activity data using wearable fitness bands / gadgets – like steps walked, time spent, minimum hours slept/rested etc.
 - c. Monitoring vital parameters on an ongoing basis
- ii) product design and premium payment frequency
 - a. One time premium using premium reset every year or
 - b. pay-as-you-go premiums over the year with more frequent resets based on actual activity monitoring
 - c. Incorporate discounts and loading based on actual reading and activity levels
- iii) Pricing approach

- a. Initial pricing be made based on standard rating factors like age, gender, Sum Insured, illness history, substance abuse (tobacco, alcohol) and basic underwriting parameters or
 - b. Standard pricing initially with pricing correction based on member activity
 - c. Ongoing pricing resets based on analysis of data
- iv) General operation
 - a. Use of standardised fitness gadgets with proper unique identification
 - b. Pricing to consider fitness gadgets or cost borne by insured?
 - c. Infrastructure to access health records of the policyholder and constant exchange of member activity and health data
- v) Servicing
 - a. Pricing should include allowance for servicing aspects like data collection, storage, retrieval, analysis and usage
 - b. Include incentives for wellness initiatives to ensure the product is tailor made to the policyholder's lifestyle and needs
 - c. Pricing should also include costs for:
 - i. Management expenses
 - ii. Device related costs and expenses
 - iii. Margins

[8]

b)

- Regulations
- Reinsurance
- Expenses
- Competition
- Global trends / practices
- Market acceptance

[2]

iii)**DEMAND SIDE (POLICYHOLDER):****Advantages:**

1. Cost Savings:
 - a. Arising from self – service / own lifestyle based incentives and discounts
 - b. Reduced acquisition costs (online sales and renewals)
 - c. Customised premium rate and not an average 'pooled' rate
2. Better lifestyles:
 - a. Lesser ailments through encouraged wellness initiatives
 - b. Better monitoring of health parameters
3. Transparency and increased trust:
 - a. Better knowledge and control over the rating approach distinguished individually

- b. Reduced risk of litigation owing to increased trust and understanding
- 4. Health management:
 - a. Proactive insurer – policyholder interactions would lead to early indicators of changing lifestyles and impact on long term health
 - b. Early assistance through early detection of potential health hazards.

Challenges in Indian context:

1. Data Privacy:
 - a. Micro management of policyholder behaviour could lead to a sense of ‘being watched’ – Data Protection and Security laws
 - b. Threat of misuse / abuse of individual data – who owns it? Who’s using it?
 - c. Consented sharing of personal data could lead to weaker enforceability in courts
2. Possible hidden costs:
 - a. Though the policyholder is expected to gain from apparent reduction in costs, there could be significant hidden costs towards infrastructure for data collection / analytics and usage
 - b. The costs with respect to the wearable devices could be transferred to the policyholder which could considerably increase the cost for the customer
3. Installation issues:
 - a. The data sent from the device is as good as the device itself and faulty devices could have a significant bearing on the potential premiums payable
 - b. Technical issues and constant servicing issues of the device could further inflate the costs of the end user.
4. Genetics and claimant behaviour:
 - a. Premiums based on standard readings on devices could ignore the genetic make-up of individuals pertaining to particular genetic heritage.
 - b. For example, the initial premiums could be the same for a traditional product or a UBI product for a policyholder having a worse than normal reading of, say, blood pressure. However, in case of a claim free year, the former is likely to get a NCB on his / her Sum Insured whereas the latter could still continue to pay higher premiums because of the ‘worse than normal’ reading.
5. Accessibility and Insurance for all:
 - a. Most economically weaker sections may get excluded unless reinforced by regulation

SUPPLY SIDE (INSURER)

Advantages:

1. Reduced risk of selection
 - a. In UBI because the premiums are estimated at an individual level, there is reduced risk of self-selection particularly at renewals
 - b. In fact, the insurers could benefit from a predominant positive selection i.e. those policyholders who are already leading a healthy lifestyle are more likely to voluntarily purchase UBI products.

2. Better claim experience:
 - a. A corollary to positive selection is better claim experience for the insurer arising from insuring 'better lives'.
 - b. Better claim management through early detection and proactive measures for claim prevention
 - c. More customised approach towards identifying the types of risk to avoid leading to better claims experience
3. Improved Customer service:
 - a. Improved business intelligence leading to product innovation and tailor made products
 - b. Attracting best customer through incentives and initiatives
 - c. Expedited claim processing (real time notifications, online monitoring and faster payments)
 - d. Increased end user satisfaction through better CRM
4. Reduced overheads:
 - a. More efficient policy issuance and claim processing
 - b. Better control on business arising from better understanding of customers

CHALLENGES IN INDIAN CONTEXT

1. Basic tenet of insurance at risk:
 - a. Insurance is a concept of pooling wherein not all risks operate the same way. However customised products converts insurance into a pure financial service rather than insurance
 - b. Significant variations in premiums across end users arising from customization could lead to lesser transparency and hence makes policy administration more complex
2. Regulations
 - a. Considering the apparent price discrimination, regulation may disallow usage based insurance products in health
 - b. Besides, health insurance being a necessary social security, pooling of risks ensures a larger section of the society is covered whereas UBI could lead to unreasonably high premiums for some leading to them being left uncovered.
 - c. UBI may not fit into guaranteed renewals and health insurance for all, one of the important tenets of Health insurance regulations
 - d. EoM caps may not be met if each policy is priced in a tailor made fashion
3. Insurer's capabilities
 - a. Most important is the capability to granularly collect and analyse data
 - b. Technology and infrastructure required to enable functioning of UBI product
 - c. Significant investment costs which could render the premium unreasonably high
 - d. Capability to manage cyber security threats
 - e. Exposure to liability risks on using private data
 - f. Reinsurance availability or setting up a reinsurance captive
4. Frauds:
 - a. Wearable technology is prone to fraud / abuse
 - b. Unique identification of gadgets requires technology for tracking and use

- c. Intended use of gadget could itself be questionable
5. Extreme dependence on technology
 - a. GIGO – garbage in garbage out
 - b. Operational risks – device failures, system failures, lost connectivity, location / geographical constraints and so on
6. Demand side constraints
 - a. At a time where there is still lot to be desired in health insurance penetration in India, having supply side 'selection' of risk could lead to wider gap between the insurable and the insured
 - b. Besides, there could be a significant number of people who are left out in this era of digital revolution (like rural population). It is important to focus on inclusive insurance before going for exclusive insurance products.

[20]

[36 Marks]**Solution 5:****i)**

1. Change in internal risk management framework and ideology
2. Decision to transfer risk than to retain it
3. Change in the management (owing to merger / buy out) leading to change in the risk management outlook
4. Regulatory requirements
5. Recognising that the same be handled by experts
6. Large capital requirements (regulatory / solvency) to support the captive
7. In some situations the parent may decide only to derisk a captive i.e. exit from old years only but not to wholly close out the captive.

[4]

ii)

Potential benefits:

- Insurance liabilities by their nature may take a considerable time to be reported, and even longer to settle, particularly long tailed. The insurer would be free of the legal, regulatory and other uncertainties arising out of liability line
- An exit strategy can help the insurer exit the inherent volatility and uncertainty in estimation of premiums and reserves.
- Retaining correlated risks could mean more capital requirements than that in transferring risks. Therefore an exit could potentially free up capital for the insurer.
- A captive could itself have a minimum capital requirement irrespective of its size. This could have an onerous effect on the capital of the parent company.
- The redundancy of an insurer within an insurer could mean significant costs of set up and management of the captive. These expenses could be significantly reduced.

- A captive could have its own set of regulatory, governance and compliance requirements like annual actuarial reviews, board of director meetings, regular audits, etc.
- Soft markets could mean better premium rates and hence it makes better commercial sense to transfer the risk to market insurers
- A captive is a mode of self-insurance and does not offer a diversification benefit in assessing solvency capital requirements. Exiting and purchasing cover from the market could help relieve solvency requirements.
- A captive's capital may be subject to investment restrictions applicable for insurance companies. Exiting can expose the free reserves of the captive to better investment avenues
- The insurer can benefit from market expertise in getting D&O insurance

[8]

[12 Marks]
